

Tut 4a–4c: Partial Derivatives (3 hours)

UECM1304 Calculus II

Remark: The answers are for you to check your working. There may be errors. Please report to the lecturer if you found any errors. If you are not familiar with the workings, you may ask a good AI assistant online. Make sure to check the workings of AI since AI can also make mistakes.

4a(i) Multivariate Functions

1. Let $f(x, y) = \ln(x + y - 1)$.
 - (a) Evaluate $f(1, 1)$ and $f(e, 1)$.
 - (b) Find and sketch the domain of f .
 - (c) Find the range of f .

Ans:

```
(define (f x y) (log (- (+ x y) 1)))  
(display (f 1 1))  
(display (f (exp 1) 1))
```

- (a) 0, 1
- (b) $\{(x, y) : x + y > 1\}$
- (c) $\mathbb{R}$

2. Let $f(x, y, z) = e^{\sqrt{z-x^2-y^2}}$.
 - (a) Evaluate $f(2, -1, 6)$.
 - (b) Find the domain of f .
 - (c) Find the range of f .

Ans:

```
(define (f x y z) (exp (sqrt (- z (* x x) (* y y)))))  
(display (f 2 -1 6))
```

- (a) e
- (b) $\{(x, y, z) : x^2 + y^2 \leq z\}$
- (c) $[0, \infty)$

3. Find and sketch the domain of the function.
 - (a) $f(x, y) = \ln(9 - x^2 - 9y^2)$
 - (b) $f(x, y, z) = \sqrt{1 - x^2 - y^2 - z^2}$

Ans: (a) $\mathcal{D} = \{(x, y) \in \mathbb{R}^2 : x^2 + 9y^2 < 9\}$
 (b) $\mathcal{D} = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$

4. Draw a contour map of the function $f(x, y) = (y - 2x)^2$ showing several level curves.

Ans: Using <https://www.tools-online.app/tools/gnuplot>

```
set contour
set cntrparam levels 20
splot [-5:5] [-5:5] (y-2*x)**2    # y = 2x + c
```

Note: Working with GeoGebra classic is possible but tedious

4a(ii) Limit and Continuity

5. Find the limit, if it exists, or show that the limit does not exist.

(a) $\lim_{(x,y) \rightarrow (1,0)} \ln\left(\frac{1+y^2}{x^2+xy}\right)$ [Ans: Exists]

(b) $\lim_{(x,y) \rightarrow (0,0)} \frac{y^4}{x^4 + 3y^4}$ [Ans: No]

(c) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin^2 y}{x^2 + 2y^2}$ [Ans: No]

(d) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 1} - 1}$ [Ans: Exists]

(e) $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{x^2 + 2y^2 + 3z^2}{x^2 + y^2 + z^2}$ [Ans: No]

6. Determine the set of points at which the function is continuous.

(a) $f(x, y) = \begin{cases} \frac{x^2 y^3}{2x^2 + y^2}, & \text{if } (x, y) \neq (0, 0); \\ 1, & \text{if } (x, y) = (0, 0). \end{cases}$ [Ans: $(\mathbb{R}^2)^*$]

(b) $f(x, y, z) = \frac{\sqrt{y}}{x^2 - y^2 + z^2}$ [Ans: $\{(x, y, z) \in \mathbb{R}^3 : y^2 \neq x^2 + z^2\}$]

7. Show that the function f given by $f(\mathbf{x}) = \|\mathbf{x}\|$ is continuous on $\mathbb{R}^n$. [Hint: Use triangle inequality]

4a(iii) Partial Derivatives

8. Find the first partial derivatives of the function.

(a) $f(x, y) = xe^{3y}$

(b) $f(x, y) = \int_y^x \cos(t^2) dt$

(c) $w = \ln(x + 2y + 3z)$

Ans:

(a) $f_x = e^{3y}$, $f_y = 3xe^{3y}$

(b) $f_x = \cos(x^2)$, $f_y = -\cos(y^2)$

(c) $w_x = \frac{1}{x+2y+3z}$, $w_y = \frac{2}{x+2y+3z}$, $w_z = \frac{3}{x+2y+3z}$

9. Find the indicated partial derivative.

(a) $f(x, y) = 3xy^4 + x^3y^2$; f_{xxy} , f_{yyy} [Ans: $12xy$, $72xy$]

(b) $u = e^{r\theta} \sin \theta$; $\frac{\partial^3 u}{\partial r^2 \partial \theta}$ [Ans: $e^{r\theta} r \theta^2 \sin \theta + 2e^{r\theta} \theta \sin \theta + e^{r\theta} \theta^2 \cos \theta$]

(c) $w = \frac{x}{y+2z}$; $\frac{\partial^3 w}{\partial z \partial y \partial x}$, $\frac{\partial^3 w}{\partial x^2 \partial y}$ [Ans: $4(y+2z)^{-3}$, $w_{yxx} = 0$]

10. If $u = e^{a_1 x_1 + a_2 x_2 + \dots + a_n x_n}$, where $a_1^2 + a_2^2 + \dots + a_n^2 = 1$, show that

$$\frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} + \dots + \frac{\partial^2 u}{\partial x_n^2} = u$$

4b(i) Directional Derivatives and Gradient Vectors

11. Find the directional derivative of the function at the given point in the direction of the vector $\vec{v}$.

(a) $f(x, y) = 1 + 2x\sqrt{y}$, $(3, 4)$, $\vec{v} = (4, -3)$ [Ans: $23/10$]

(b) $f(x, y, z) = xe^y + ye^z + ze^x$, $(0, 0, 0)$, $\vec{v} = (5, 1, -2)$ [Ans: $4/\sqrt{30}$]

(c) $g(x, y, z) = (x + 2y + 3z)^{3/2}$, $(1, 1, 2)$, $\vec{v} = (0, 2, -1)$ [Ans: $9/\sqrt{20}$]

12. Find the maximum rate of change of f at the given point and the direction in which it occurs.

(a) $f(x, y) = \frac{y^2}{x}$, $(2, 4)$ [Ans: $4\sqrt{2}$, $(-1/\sqrt{2}, 1/\sqrt{2})$]

(b) $f(x, y, z) = \frac{x+y}{z}$, $(1, 1, -1)$ [Ans: 1 , $(0, 1)$]

4b(ii) Tangent Planes and Linear Approximations

13. Find an equation of the tangent plane to the given surface at the specified point:

$$z = e^{x^2 - y^2}, \quad (1, -1, 1).$$

[Ans: $z - 1 = 2(x - 1) + 2(y + 1)$]

14. Explain why the function is differentiable at the given point. Then find the linearisation $L(x, y)$ of the function at that point.

(a) $f(x, y) = x\sqrt{y}$, $(1, 4)$ [Ans: $2x + y/4 - 1$]

(b) $f(x, y) = e^{-xy} \cos y$, $(\pi, 0)$ [Ans: $1 - \pi y$]

15. Find the linear approximation of the function $f(x, y) = \sqrt{20 - x^2 - 7y^2}$ at $(2, 1)$ and use it to approximate $f(1.95, 1.08)$. [Ans: $(20 - 2x - 7y)/3$]

4b(iii) Total Differential

16. Find the differential of the function.

(a) $z = x^3 \ln(y^2)$ [Ans: $6x^3 \ln y dx + 2x^3 dy/y$]

(b) $m = p^5 q^3$ [Ans: $5p^4 q^3 dp + 3p^5 q^2 dq$]

17. If $z = 5x^2 + y^2$ and (x, y) changes from $(1, 2)$ to $(1.05, 2.1)$, compare the values of Δz and dz . [Ans: $dz = 0.9$, $\Delta z = (5(1.05)^2 + 2.1^2 - 5(1)^2 - 2^2) = 0.9225$]

18. A boundary stripe 8 cm wide is painted around a rectangle whose dimensions are 30 m by 60 m. Use differentials to approximate the number of square meters of paint in the stripe. [Ans: 14.4]

4c(i) Chain Rule (and Tree Diagram)

19. Use the Chain Rule to find dz/dt and dw/dt .

(a) $z = \sin x \cos y$, $x = \pi t$, $y = \sqrt{t}$ [Ans: $\pi \cos(\pi t) \cos \sqrt{t} - \frac{1}{2}x^{-1/2} \sin(\pi t) \sin \sqrt{t}$]

(b) $w = xe^{y/z}$, $x = t^2$, $y = 1 - t$, $z = 1 + 2t$ [Ans: $e^{y/z}(2t - x/z - 2xy/z^2)$]

20. Suppose f is a differentiable function of x and y , and $g(u, v) = f(e^u + \sin v, e^u + \cos v)$. Use the table of values to calculate $g_u(0, 0)$ and $g_v(0, 0)$.

	f	g	f_x	f_y
$(0, 0)$	3	6	4	8
$(1, 2)$	6	3	2	5

[Ans: 7, 2]

21. Use a tree diagram to write out derivatives w.r.t. x and y using Chain Rule for $w = f(r, s, t)$, where $r = r(x, y)$, $s = s(x, y)$, $t = t(x, y)$. Assume all functions are differentiable.

22. Let $M = xe^{y-z^2}$, $x = 2uv$, $y = u - v$, $z = u + v$. Use the Chain Rule to find the partial derivatives $\frac{\partial M}{\partial u}$ and $\frac{\partial M}{\partial v}$ when $u = 3$, $v = -1$. [Ans: 16, 36]

23. Find dy/dx .

(a) $\sqrt{xy} = 1 + x^2y$ [Ans: $-(4xy\sqrt{xy} - y)/(2x^2\sqrt{xy} - x)$]

(b) $\cos(x - y) = xe^y$ [Ans: $(\sin(y - x) - e^y)/(\sin(y - x) + xe^y)$]

24. Assume that all the given functions are differentiable. If $z = f(x, y)$, where $x = s + t$ and $y = s - t$, show that

$$\left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2 = \frac{\partial z}{\partial s} \frac{\partial z}{\partial t}.$$

4c(ii) Implicit Differentiation

25. Use implicit differentiation to find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

(a) $x^2 + y^2 + z^2 = 3xyz$ [Ans: $(3zy - 2x)/(2z - 3xy)$, $(2y - 3xz)/(3xy - 2z)$]

(b) $\sin(xyz) = x + 2y + 3z$ [Ans: $(1 - yz \cos(xyz))/(xy \cos(xyz) - 3)$, $(2 - xz \cos(xyz))/(xy \cos(xyz) - 3)$]

4c(iii) Maximum and Minimum Values

26. Find the local maximum and minimum values and saddle point(s) of the function.

(a) $f(x, y) = 9 - 2x + 4y - x^2 - 4y^2$ [Ans: max $(-1, 1/2)$]

(b) $f(x, y) = x^2 + y^2 + x^2y + 4$ [Ans: max $(0, 0)$, saddles $(\pm\sqrt{2}, -1)$]

27. Find the absolute maximum and minimum values of f on the set $\mathcal{D}$.

(a) $f(x, y) = 1 + 4x - 5y$, $\mathcal{D}$ is the closed triangular region with vertices $(0, 0)$, $(2, 0)$ and $(0, 3)$. [Ans: max $(2, 0)$; min $(0, 3)$]

(b) $f(x, y) = x^2 + y^2 + x^2y + 4$, $\mathcal{D} = \{(x, y) | |x| \leq 1, |y| \leq 1\}$. [Ans: max $(\pm 1, 1)$; min $(0, 0)$]

(c) $f(x, y) = 2x^3 + y^4$, $\mathcal{D} = \{(x, y) | x^2 + y^2 \leq 1\}$. [Ans: max $(1, 0)$; min $(-1, 0)$]

28. Find the extreme values of $f(x, y) = 2x^2 + 3y^2 - 4x - 5$ on the region $x^2 + y^2 \leq 16$.

[Ans: Inside: min: $(1, 0)$; Boundary: max: $(-1/3, \sqrt{143}/3)$]

4c(iv) Lagrange Multipliers (Method)

29. Find three positive numbers whose sum is 12 and the sum of whose squares is as small as possible. [Ans: $x = y = z$]

30. Use Lagrange multipliers to find the maximum and minimum values of the function subject to the given constraint.

(a) $f(x, y) = xy$; $4x^2 + y^2 = 8$
[Ans: $(x, y, \lambda) = (\pm 1, \pm 2, -1/4)$ (max); $(x, y, \lambda) = (\pm 1, \mp 2, 1/4)$ (min)]

(b) $f(x, y, z) = 2x + 6y + 10z$; $x^2 + y^2 + z^2 = 35$
[Ans: $(x, y, z, \lambda) = (1, 3, 5, -1)$ (max); $(x, y, z, \lambda) = (-1, -3, -5, 1)$ (min)]

(c) $f(x, y, z) = x + 2y$; $x + y + z = 1$, $y^2 + z^2 = 4$
[Ans: $(x, y, z, \lambda_1, \lambda_2) = (1, -\sqrt{2}, \sqrt{2}, -1, 1/2^{3/2})$ (min);
 $(x, y, z, \lambda_1, \lambda_2) = (1, \sqrt{2}, -\sqrt{2}, -1, -1/2^{3/2})$ (max)]